

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Buffer concentrate pH 6.00**  
 Product Description: **Buffer concentrate pH 6.00**  
 Cat No. : SB106-500  
 Synonyms: None

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use: Laboratory chemicals.  
 Uses advised against: No Information available

**Company** Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                 |                    |
|-----------------|--------------------|
| Carcinogenicity | Category 1B (H350) |
|-----------------|--------------------|

**Label Elements**

**Signal Word**
**Danger**
**Hazard Statements**

H350 - May cause cancer

**Precautionary Statements**  
**Prevention**

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P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P308 + P313 - IF exposed or concerned: Get medical advice/attention

## Storage

P403 - Store in a well-ventilated place

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                   | CAS No    | Weight % |
|-----------------------------|-----------|----------|
| Water                       | 7732-18-5 | 91.02    |
| Sodium phosphate, monobasic | 7558-80-7 | 8        |
| Sodium hydroxide            | 1310-73-2 | 0.7      |
| Formaldehyde                | 50-00-0   | 0.2      |
| Methyl alcohol              | 67-56-1   | 0.08     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Use personal protective equipment as required.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

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## Extinguishing media

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

## Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

## **Hazardous Combustion Products**

None under normal use conditions.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component        | Malaysia | ACGIH TLV                    | OSHA PEL                                                           |
|------------------|----------|------------------------------|--------------------------------------------------------------------|
| Sodium hydroxide |          | Ceiling: 2 mg/m <sup>3</sup> | (Vacated) Ceiling: 2 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup> |

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|                |  |                                       |                                                                                                                                                                                          |
|----------------|--|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formaldehyde   |  | TWA: 0.1 ppm<br>STEL: 0.3 ppm         | (Vacated) TWA: 3 ppm<br>(Vacated) STEL: 10 ppm<br>(Vacated) Ceiling: 5 ppm<br>TWA: 0.75 ppm<br>STEL: 2 ppm                                                                               |
| Methyl alcohol |  | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 260 mg/m <sup>3</sup><br>(Vacated) STEL: 250 ppm<br>(Vacated) STEL: 325 mg/m <sup>3</sup><br>Skin<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> |

| Component        | European Union                                                                                                                                                                    | The United Kingdom                                                                                                      | Germany                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sodium hydroxide |                                                                                                                                                                                   | 2 mg/m <sup>3</sup> STEL                                                                                                | 2 mg/m <sup>3</sup> TWA (inhalable fraction)                                                                                                                                                                                                                                                                                                                        |
| Formaldehyde     | TWA: 0.37 mg/m <sup>3</sup> (8h)<br>TWA: 0.62 mg/m <sup>3</sup> (8h)<br>TWA: 0.3 ppm (8h)<br>TWA: 0.5 ppm (8h)<br>Skin<br>STEL: 0.74 mg/m <sup>3</sup> (8h)<br>STEL: 0.6 ppm (8h) | STEL: 2 ppm 15 min<br>STEL: 2.5 mg/m <sup>3</sup> 15 min<br>TWA: 2 ppm 8 hr<br>TWA: 2.5 mg/m <sup>3</sup> 8 hr<br>Carc. | TWA: 0.3 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 0.37 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 0.3 ppm (8 Stunden). MAK no irritation should occur during mixed exposure<br>TWA: 0.37 mg/m <sup>3</sup> (8 Stunden). MAK no irritation should occur during mixed exposure<br>Höhepunkt: 0.6 ppm<br>Höhepunkt: 0.74 mg/m <sup>3</sup> |
| Methyl alcohol   | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr<br>Skin                                                                                                                      | WEL - TWA: 200 ppm TWA; 266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm STEL; 333 mg/m <sup>3</sup> STEL               | 100 ppm TWA MAK; 130 mg/m <sup>3</sup> TWA MAKSkin absorber                                                                                                                                                                                                                                                                                                         |

## Exposure Controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

### Eye Protection

Wear safety glasses with side shields (or goggles)

### Hand Protection

Protective gloves

### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

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Environmental exposure controls No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Appearance     | No information available |
| Physical State | Liquid                   |
| Odor           | Odorless                 |
| Odor Threshold | No data available        |
| pH             | 6.0                      |

|                     |                   |
|---------------------|-------------------|
| Melting Point/Range | 0 °C / 32 °F      |
| Softening Point     | No data available |
| Boiling Point/Range | 100 °C / 212 °F   |
| Flash Point         | Not applicable    |

Method - No information available

|                          |                          |
|--------------------------|--------------------------|
| Evaporation Rate         | No information available |
| Flammability (solid,gas) | Not applicable           |
| Explosion Limits         | No data available        |

Liquid

|                              |                          |
|------------------------------|--------------------------|
| Vapor Pressure               | No information available |
| Vapor Density                | No information available |
| Specific Gravity / Density   | No data available        |
| Bulk Density                 | Not applicable           |
| Water Solubility             | Miscible                 |
| Solubility in other solvents | No information available |

(Air = 1.0)

Liquid

### Partition Coefficient (n-octanol/water)

| Component      | log Pow |
|----------------|---------|
| Formaldehyde   | -0.35   |
| Methyl alcohol | -0.74   |

|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | No data available        |
| Decomposition Temperature | No data available        |
| Viscosity                 | No data available        |
| Explosive Properties      | No information available |
| Oxidizing Properties      | No information available |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

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## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Excess heat.

## Incompatible Materials

None known.

## Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

##### **(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component                   | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|-----------------------------|--------------------------------|-------------------------------|-------------------------------|
| Water                       | -                              | -                             | -                             |
| Sodium phosphate, monobasic | LD50 = 8290 mg/kg ( Rat )      | LD50 > 7940 mg/kg ( Rabbit )  | LC50 > 0.83 mg/L ( Rat ) 4 h  |
| Sodium hydroxide            | LD50 = 325 mg/kg ( Rat )       | LD50 = 1350 mg/kg ( Rabbit )  | -                             |
| Formaldehyde                | 500 mg/kg (Rat)                | LD50 = 270 mg/kg (Rabbit)     | 0.578 mg/L (Rat) 4 h          |
| Methyl alcohol              | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |

##### **(b) skin corrosion/irritation;**

No data available

##### **(c) serious eye damage/irritation;**

No data available

##### **(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

No data available

| Component                       | Test method                                  | Test species      | Study result                |
|---------------------------------|----------------------------------------------|-------------------|-----------------------------|
| Formaldehyde<br>50-00-0 ( 0.2 ) | Skin sensitization<br>Test method Patch Test | Man<br>guinea pig | Sensitizer<br>Sensitization |
|                                 | Respiratory sensitization<br>in vitro        |                   |                             |

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|                                    |                                                                   |            |                 |
|------------------------------------|-------------------------------------------------------------------|------------|-----------------|
| Methyl alcohol<br>67-56-1 ( 0.08 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test<br>(GPMT) | guinea pig | non-sensitising |
|------------------------------------|-------------------------------------------------------------------|------------|-----------------|

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component    | EU           | UK    | Germany | IARC    |
|--------------|--------------|-------|---------|---------|
| Formaldehyde | Carc Cat. 1B | Cat 3 |         | Group 1 |

(g) reproductive toxicity; No data available

| Component                          | Test method             | Test species / Duration          | Study result              |
|------------------------------------|-------------------------|----------------------------------|---------------------------|
| Methyl alcohol<br>67-56-1 ( 0.08 ) | OECD Test Guideline 416 | Rat / Inhalation<br>2 Generation | NOAEC =<br>1.3 mg/l (air) |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component        | Freshwater Fish                                     | Water Flea                              | Freshwater Algae                                 | Microtox                                                                        |
|------------------|-----------------------------------------------------|-----------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------|
| Sodium hydroxide | LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss) | -                                       | -                                                | -                                                                               |
| Formaldehyde     | Leuciscus idus: LC50 = 15 mg/L 96h                  | EC50 = 20 mg/L 96h<br>EC50 = 2 mg/L 48h | EC50 (72h) = 4.89 mg/L (Desmodesmus subspicatus) |                                                                                 |
| Methyl alcohol   | Pimephales promelas: LC50 > 10000 mg/L 96h          | EC50 > 10000 mg/L 24h                   |                                                  | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min |

### Persistence and degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

| Component                       | Degradability                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| Formaldehyde<br>50-00-0 ( 0.2 ) | Readily biodegradable (OECD guideline 301A, 301C and 301D) under aerobic and anaerobic conditions. |

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|                                    |                                |
|------------------------------------|--------------------------------|
| Methyl alcohol<br>67-56-1 ( 0.08 ) | DT50 ~ 17.2d<br>>94% after 20d |
|------------------------------------|--------------------------------|

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component      | log Pow | Bioconcentration factor (BCF) |
|----------------|---------|-------------------------------|
| Formaldehyde   | -0.35   | No data available             |
| Methyl alcohol | -0.74   | <10 dimensionless             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**

**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**Road and Rail Transport** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component                   | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Water                       | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |
| Sodium phosphate, monobasic | 231-449-2 | X    | X   | X     | X    | X    | X     | X    | KE-31577 |
| Sodium hydroxide            | 215-185-5 | X    | X   | X     | X    | X    | X     | X    | KE-31487 |
| Formaldehyde                | 200-001-8 | X    | X   | X     | X    | X    | X     | X    | KE-17074 |
| Methyl alcohol              | 200-659-6 | X    | X   | X     | X    | X    | X     | X    | KE-23193 |

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| Component        | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Major<br>Accident Notification | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Safety<br>Report Requirements | Rotterdam Convention<br>(PIC) | Basel Convention<br>(Hazardous Waste) |
|------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Sodium hydroxide |                                                                                                    |                                                                                                   |                               | Annex I - Y35                         |
| Formaldehyde     | 5 tonne                                                                                            | 50 tonne                                                                                          |                               |                                       |
| Methyl alcohol   | 500 tonne                                                                                          | 5000 tonne                                                                                        |                               |                                       |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**